

Nathan Gomes

240-410-8952 | nathanpgomes2023@gmail.com | [Linkedin](#) | github.com/npgomes12 | Silver Spring, MD

EDUCATION

University of Maryland

Bachelor of Science in Information Science, GPA: 3.96

College Park, MD

August 2023 – May 2027

TECHNICAL SKILLS & TECHNOLOGIES

Programming Languages: Python, SQL, R, JavaScript, HTML/CSS

Data Analysis & Visualization: Tableau, Microsoft Excel, Pandas

Tools & Technologies: Active Directory, Windows Server 2019, PowerShell, Git, GitHub, VS Code, RStudio, Oracle VirtualBox, Linux, Wireshark, Cisco Packet Tracer

IT Support: PC Troubleshooting, Hardware/Software Installation, System Configuration, User Account Management

EXPERIENCE

Hack4Impact

October 2024 – December 2024

University of Maryland (10 hours/week)

College Park, MD

- Designed and deployed full-stack web applications using Python, Node.js, and PostgreSQL for nonprofit organizations, supporting 1,000+ community members.
- Collaborated in cross-functional agile teams using Git, code reviews, and sprint workflows to deliver production-ready features on schedule.
- Performed database validation, application testing, debugging, and troubleshooting, improving system reliability and reducing technical issues.

Data Collection Temp

January 2024 – March 2024

Leidos (18 hours/week)

McLean, VA

- Captured, structured, and validated large-scale roadway test data that supported machine-learning models for real-time intersection hazard detection in the Intersection Safety Challenge (ISC).
- Troubleshoot and optimized Linux-based data acquisition systems, improving data reliability and minimizing collection downtime during field testing.
- Organized and verified large-scale roadway test data, ensuring high-quality datasets for machine-learning model development.

PROJECTS

Active Directory Lab | Windows Server 2019, PowerShell, Oracle VirtualBox

May 2026 – June 2026

- Built an Active Directory lab using Windows Server 2019 and Oracle VirtualBox to simulate an enterprise Windows environment.
- Created and managed user accounts, groups, organizational units (OUs), and permissions to support centralized identity management.
- Developed PowerShell scripts that automated the creation and management of 1,000+ Active Directory user accounts.
- Configured DNS, DHCP, and Active Directory Domain Services (AD DS) to enable secure domain authentication and network services.
- Troubleshoot user login, network connectivity, and domain-related issues, restoring system functionality.

Transportation Data Analytics Project | Python, SQL, Pandas

March 2026 – May 2026

- Developed a Python and SQL analytics pipeline that processed 100,000+ Uber ride records and identified demand trends, peak travel times, and high-traffic locations.
- Cleaned, transformed, and visualized large datasets through exploratory analysis and interactive reporting, generating actionable insights.

Traffic Citation Analytics Database | SQL

August 2025 – December 2025

- Led a four-member team in designing and implementing a fully normalized relational database consisting of 11 tables in Third Normal Form (3NF) to manage Chicago traffic citation data.
- Developed SQL queries, views, stored procedures, and aggregate functions that analyzed citation trends by ZIP code, violation type, and issue date.
- Generated reports identifying high-frequency violation areas and average fines, supporting data-driven decision-making.
- Created ER diagrams and technical documentation to maintain database scalability and data integrity.